Experiment -1

1. Setting up of the platform

We can do C programming online or offline according to the needs. It is required for us to keep working on a Linux Environment.

Online: Cocalc.ai

Steps:

1. **Create or Open a Project**
2. **Create a C Source File**
3. **Write and Prompt Code with Cocalc.ai**
4. **Open a Terminal Frame**
5. **Compile the C Code**
6. **Run and Debug**

Offline:

Methods: Dual boot, Single boot, Bootable Pen drive, VM

Steps:

1. **Set up USB Directory Structure**
2. **Download Portable C Compiler and Editor**
3. **Write C Code on Host Machine**
4. **Compile and Execute**

2. Virtual Machine Manager (Hypervisor)

Type-1 (Bare-metal)

In virtual machine management, bare metal refers to a computer's physical hardware- such as CPU, RAM, storage devices and network interfaces- without any pre-installed operating system running on top of it.

Example: Popular BareMetal Hypervisor Software:

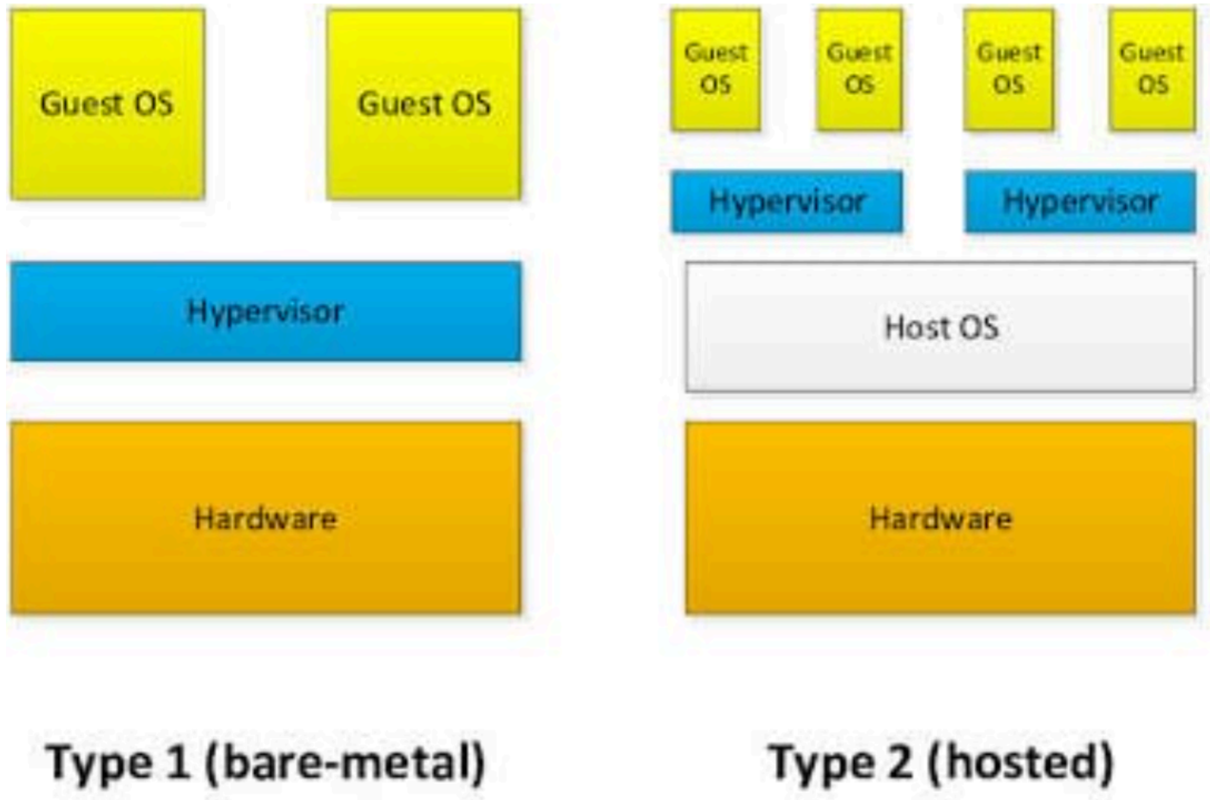
Proxmox VE & Citrix Hypervisor

Type-2 (Hosted)

A Hosted Hypervisor is virtualization software that installs as an application inside standard host operating systems (Like: Windows, Linux etc.)

Example: Oracle VirtualBox, VM Ware.

We would be using Oracle VirtualBox.



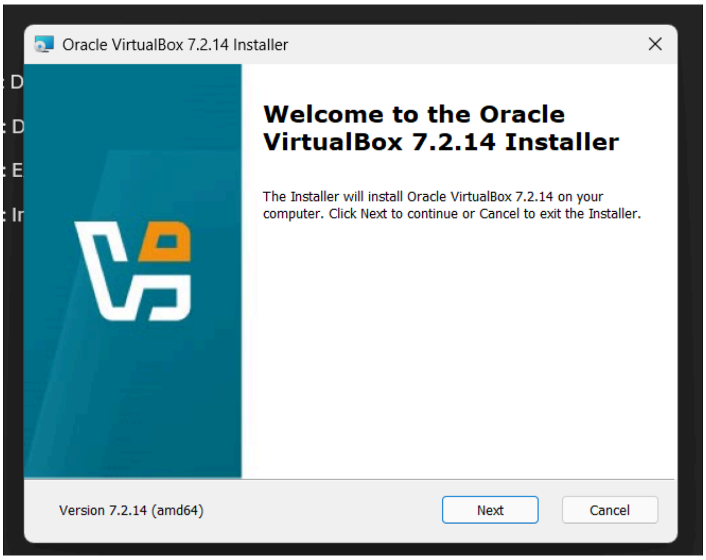
3. <u>Installation of VMM & VM</u>

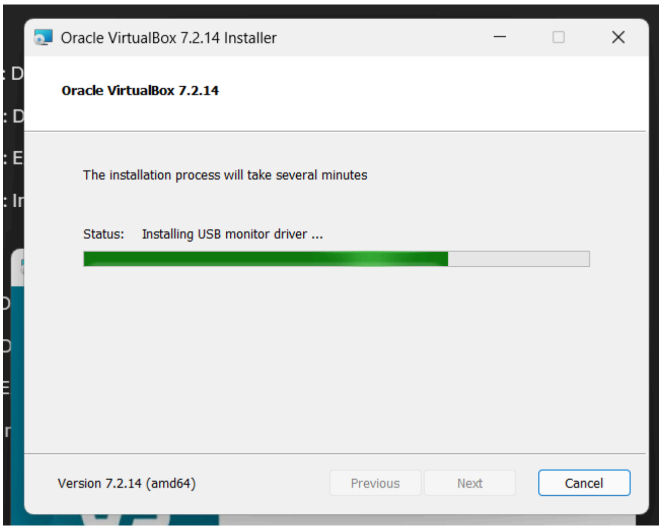
Step-1: Download Oracle Virtual Box (paste the link, attach screenshot)

Step-2: Download Kali Linux VM (paste the link, attach screenshot)

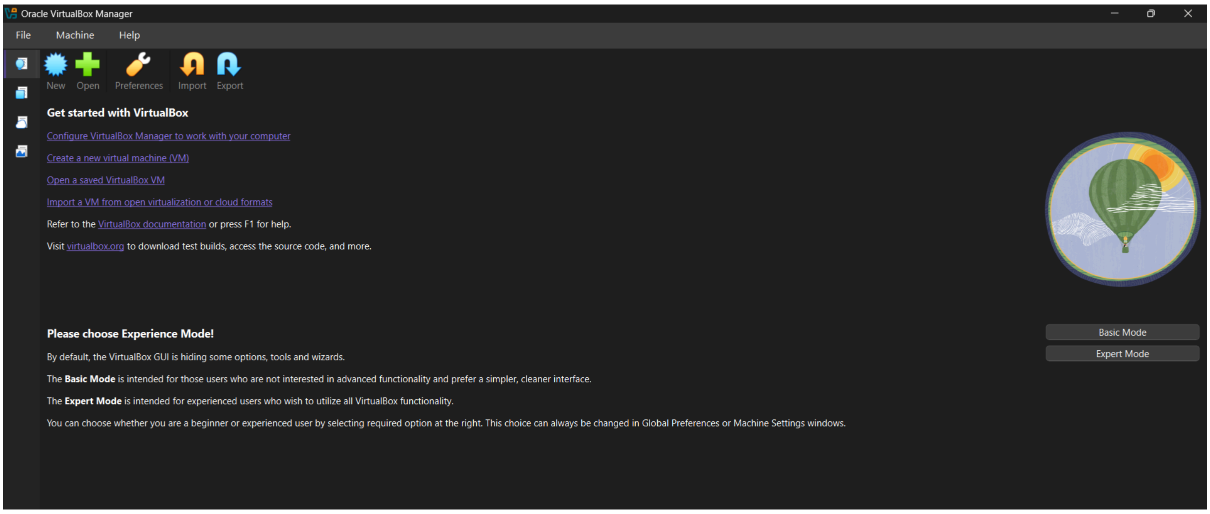
Step-3: Extracted the Kali Linux downloaded file

Step-4: Install VirtualBox

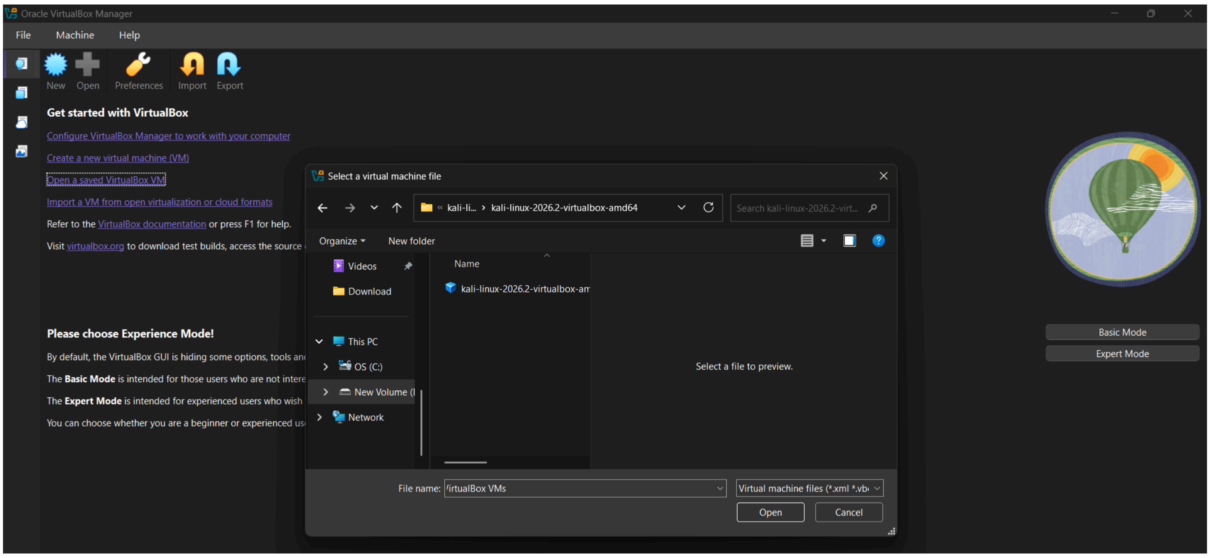


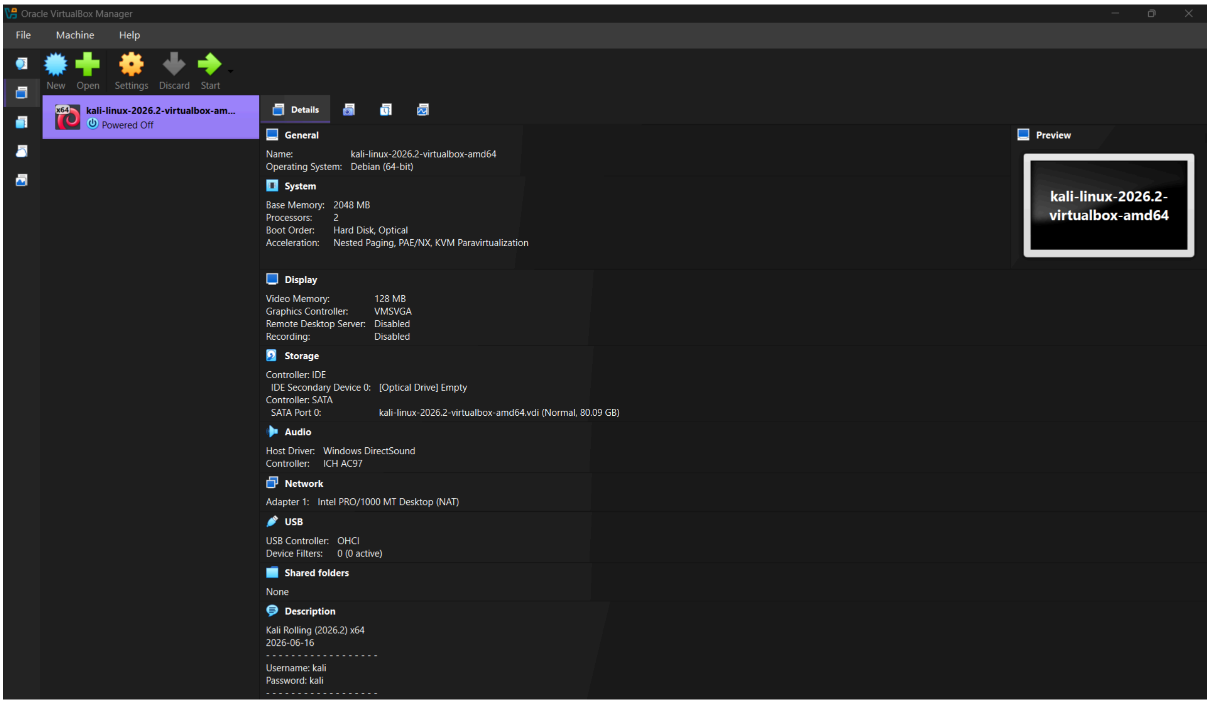


Step-5: Open Virtual box

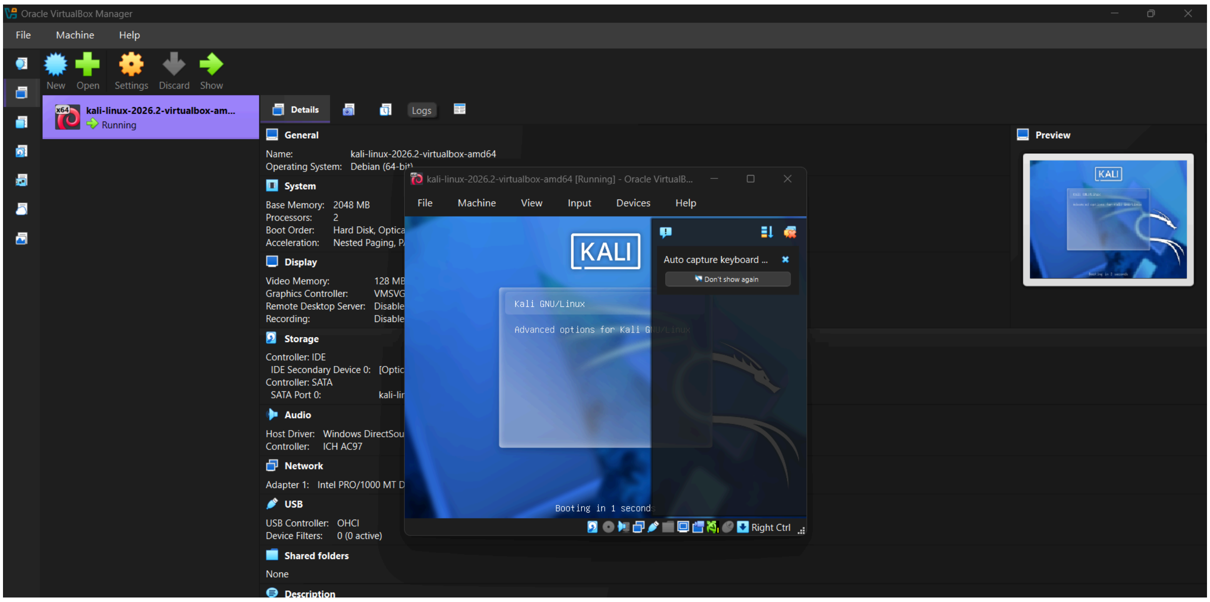


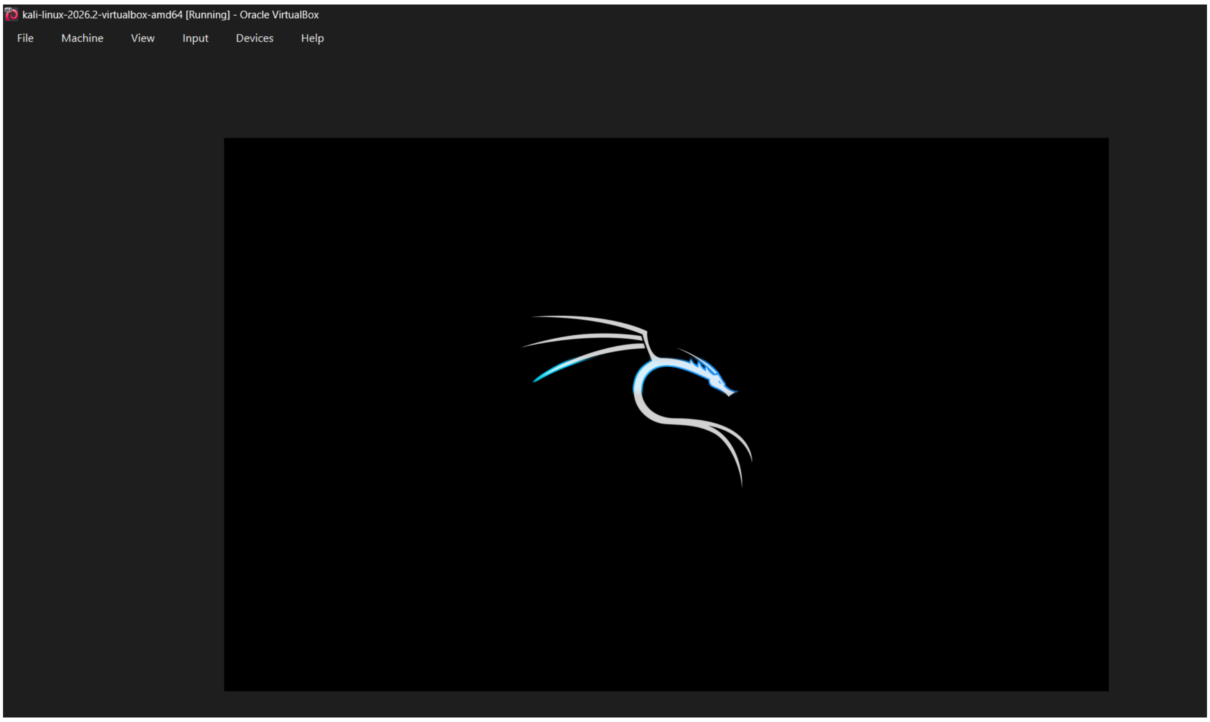
Step-6: Opening the saved VM

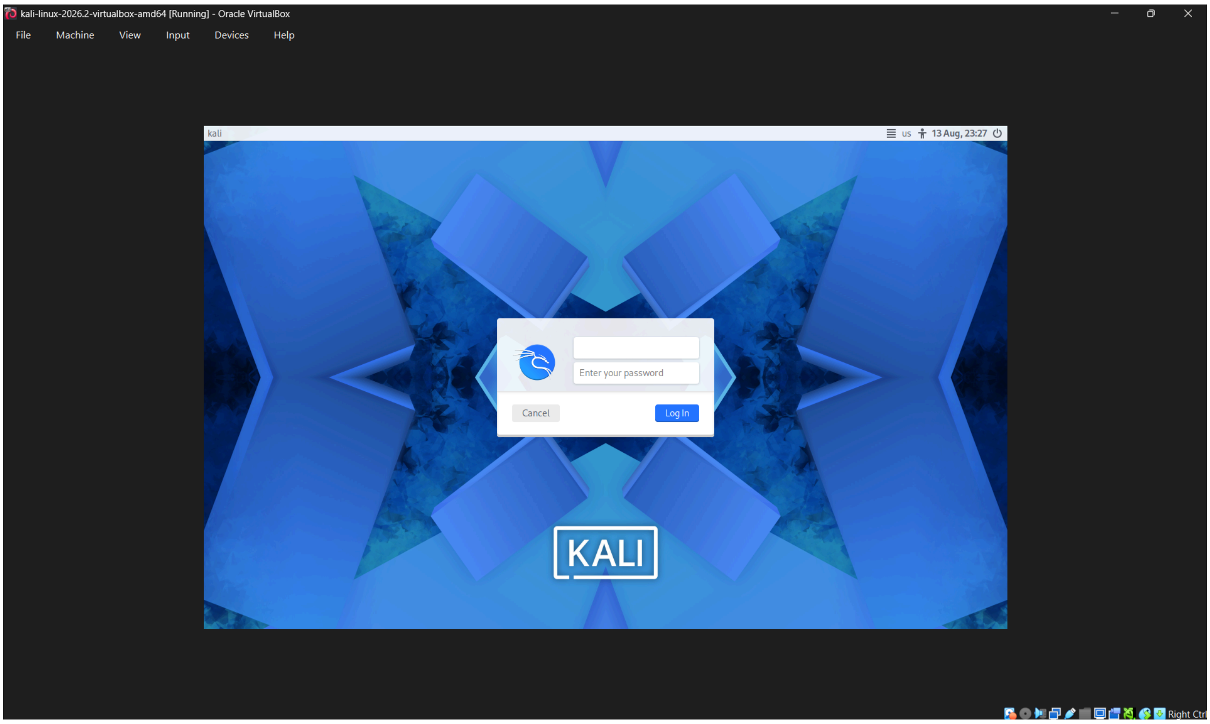




Step-7: Start the VM





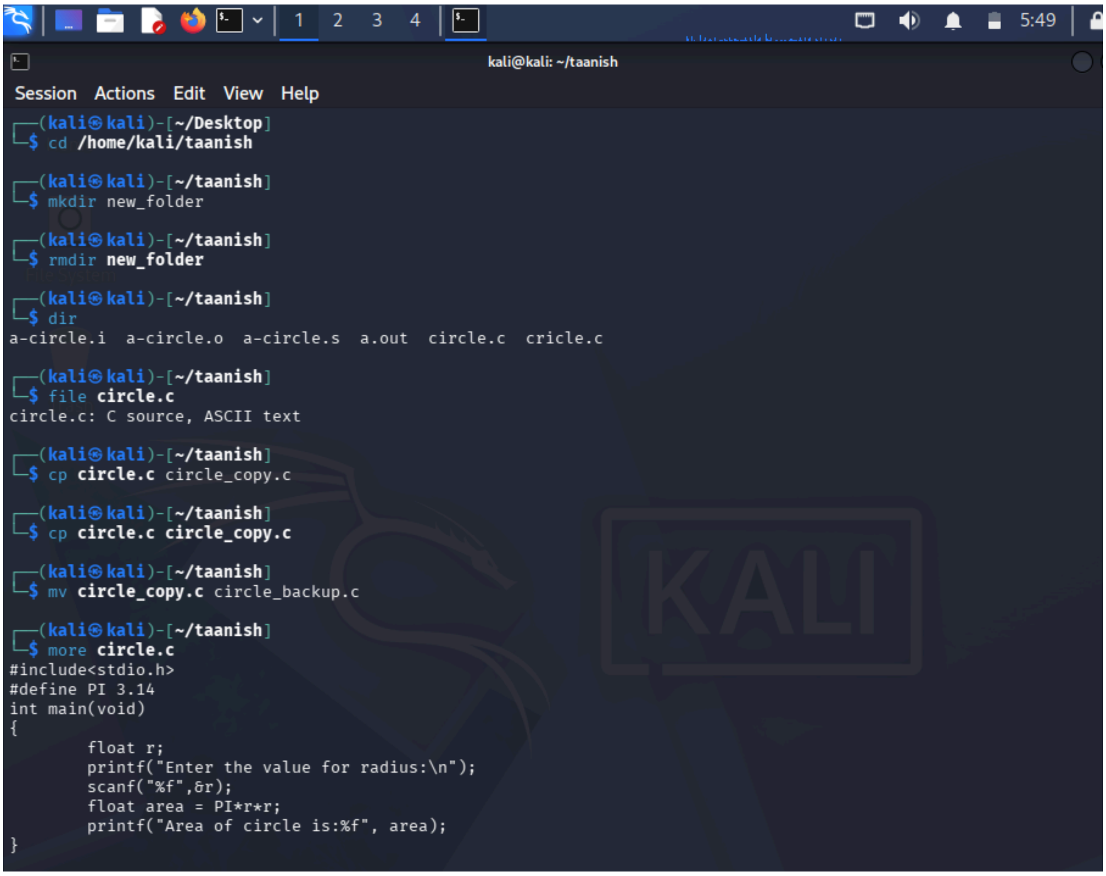


Thus, the installation of VMM (Hypervisor Type-2 Oracle VirtualBox) & VM (Kali Linux) is completed.

A) **Working with Linux Commands**

Execute the following commands in your environment, take output snippets, and provide a brief explanation.

- Terminal basics: pwd, mkdir, rmdir, dir, ls, cd, cd .., clear, history, man, whoami
- Files: vi, file, rm, cp, mv, cat, more, less, head, tail, chmod
- Redirection: >, >>, <



```
┌──(kali㉿kali)-[~/taanish]
└─$ less circle.c

┌──(kali㉿kali)-[~/taanish]
└─$ head circle.c
#include<stdio.h>
#define PI 3.14
int main(void)
{
        float r;
        printf("Enter the value for radius:\n");
        scanf("%f",&r);
        float area = PI*r*r;
        printf("Area of circle is:%f", area);
}

┌──(kali㉿kali)-[~/taanish]
└─$ tail circle.c
{
        float r;
        printf("Enter the value for radius:\n");
        scanf("%f",&r);
        float area = PI*r*r;
        printf("Area of circle is:%f", area);
}



┌──(kali㉿kali)-[~/taanish]
└─$ rm circle_backup.c

┌──(kali㉿kali)-[~/taanish]
└─$ dir
a-circle.i  a-circle.o  a-circle.s  a.out  circle.c  cricle.c

┌──(kali㉿kali)-[~/taanish]
└─$ ▮
```

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